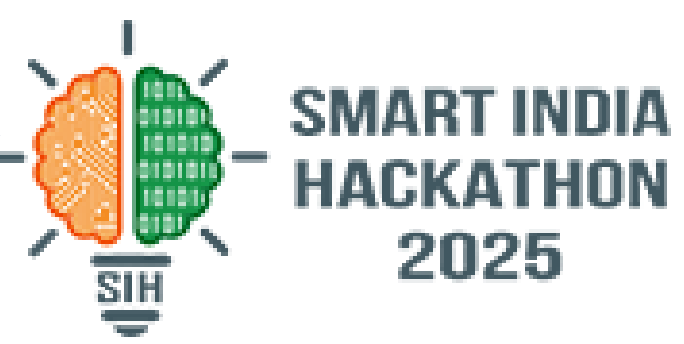


SMART INDIA HACKATHON 2025



- **Problem Statement ID** –25271
- **Problem Statement Title**-AI-Powered Import Impact Simulator for Palm Oil Tariffs
- **Theme**-Agriculture, FoodTech & Rural Development
- **PS Category**- Software
- **Team ID**-87547
- **Team Name** - Connexa



AI-Powered Agent-Based Import Impact Simulator for Palm Oil Tariffs

Current Challenges:

Over **60% palm oil import** dependency makes India's edible oil sector **vulnerable to global price shocks** and reactive policy measures, **undermining NMEO-OP's** goal of self-reliance.

Our Solution:

An **AI-powered, agent-based simulator** that **predicts how tariff changes affect** import volumes, prices, and domestic production, **empowering data-driven policy decisions.**

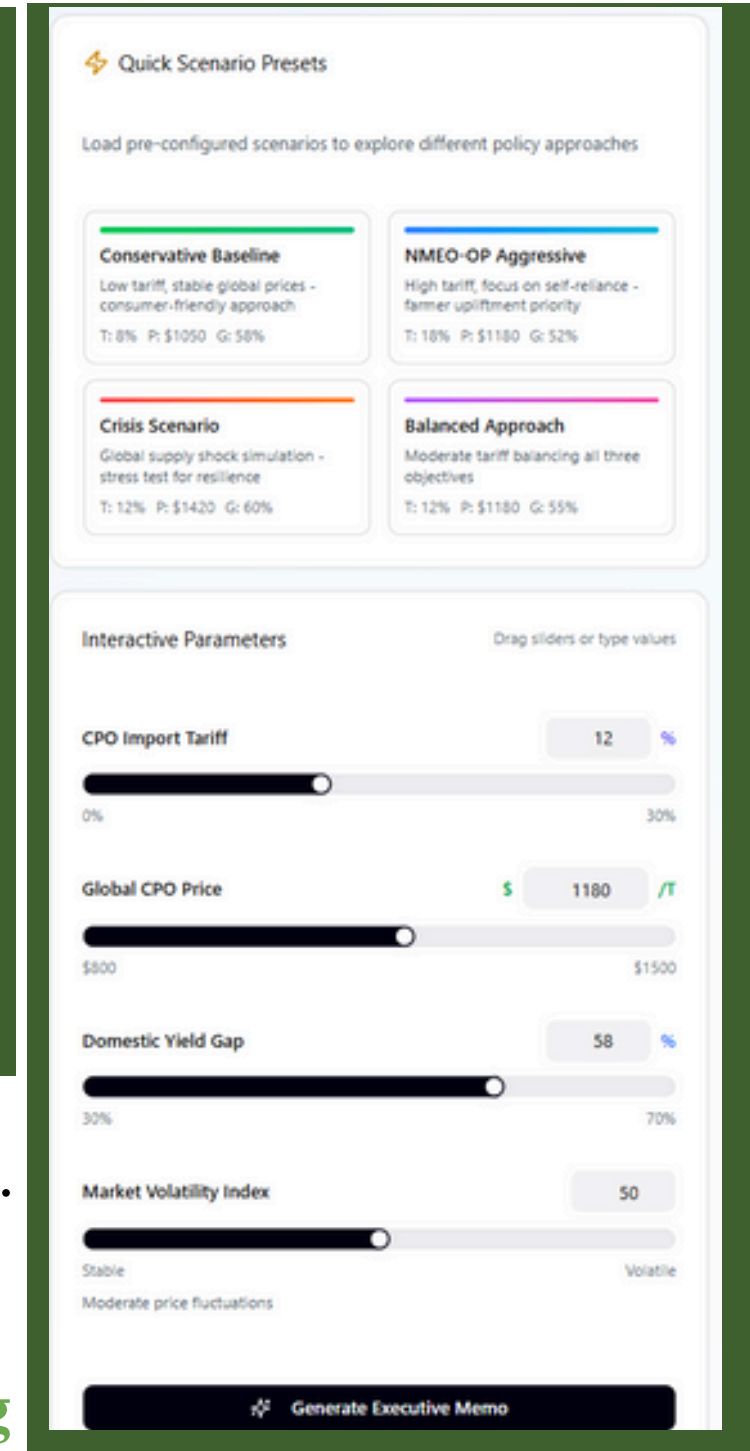
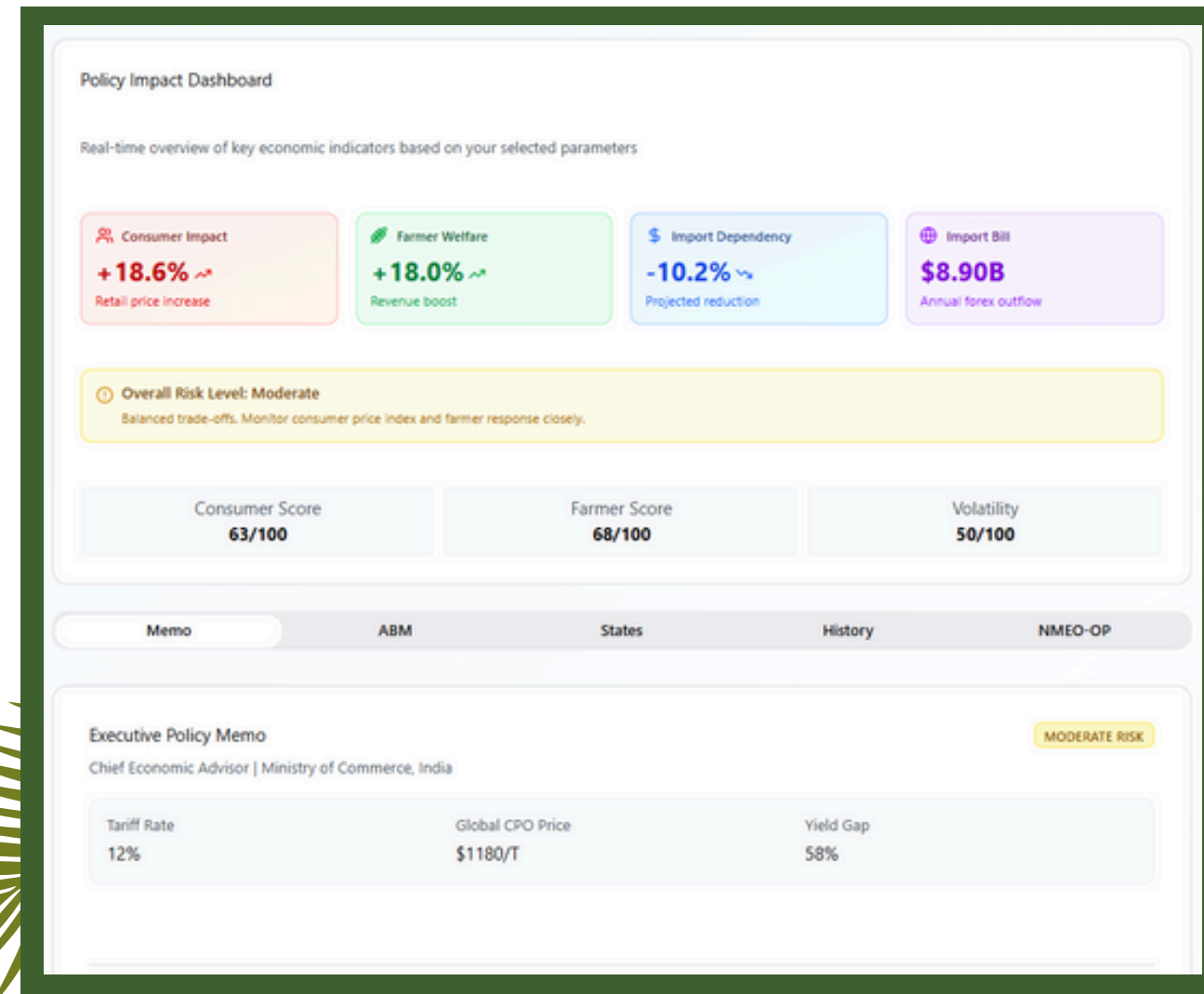
Our Unique Value Proposition :

Proactive Tariff Optimization: Quantifies **socio-economic impact** before policy implementation.

Balancing Welfare Trade-Offs: Recommends optimal tariffs to **balance farmer income** and **consumer affordability**

Real-time Decision Intelligence: Integrates **real-time global market data** with **predictive AI modeling**

Transparent Policy Tool: **Uses Explainable AI (XAI)** to justify predictions on an interactive dashboard



Data Sources :Global Commodity APIs
(Real Time CPO Price)Government Policy Data
(Tariffs, MSP)Historical Data
(Mandi, Domestic Price)**Data Ingestion and Preprocessing**Python Integration
(Data Collection and
Cleaning)Pandas
(Data Cleaning)**Data Splitting****Simulation Engine & AI Model**Time Series
(Forecasting
Prophet/
SARIMA)Agent- Based
Modelling
(ABM)
(MESA)Trained AI
(Easily
Explainable
XAI) (SHAP)**Backend & API****Interactive
Dashboard**

UI Framework

Interactive Visuals
(Charts, Sliders)Geographical
Heat Maps
(State-wise
analysis)**Frontend :** React.js, Plotly Dash**Backend :** Flask, Django REST Framework, Python
Database & Data Management : PostgreSQL,
TimescaleDB**AI/ML Models & Libraries :** Mesa (ABM), Prophet,
SARIMA, SHAP (XAI)**Integration & Data Acquisition :** APIs**MLOps & Deployment :** Docker, Git, GitHub**UI/UX :** Plotly Dash components (charts, sliders,
maps)**Prototype Link :** <https://policy-simu.vercel.app>



Feasibility

Technical: Utilizes existing technologies like **web, mobile, AI**. Models are validated using metrics such as **RMSE and MAPE** for accuracy

Economic: Low initial investment with cloud-based infrastructure. Helpful to **increase per capita income** in the field of CPO.

Operating : User-friendly interface with visuals that engage users and **PostgreSQL with Timescale DB for efficient storage**.



Viability

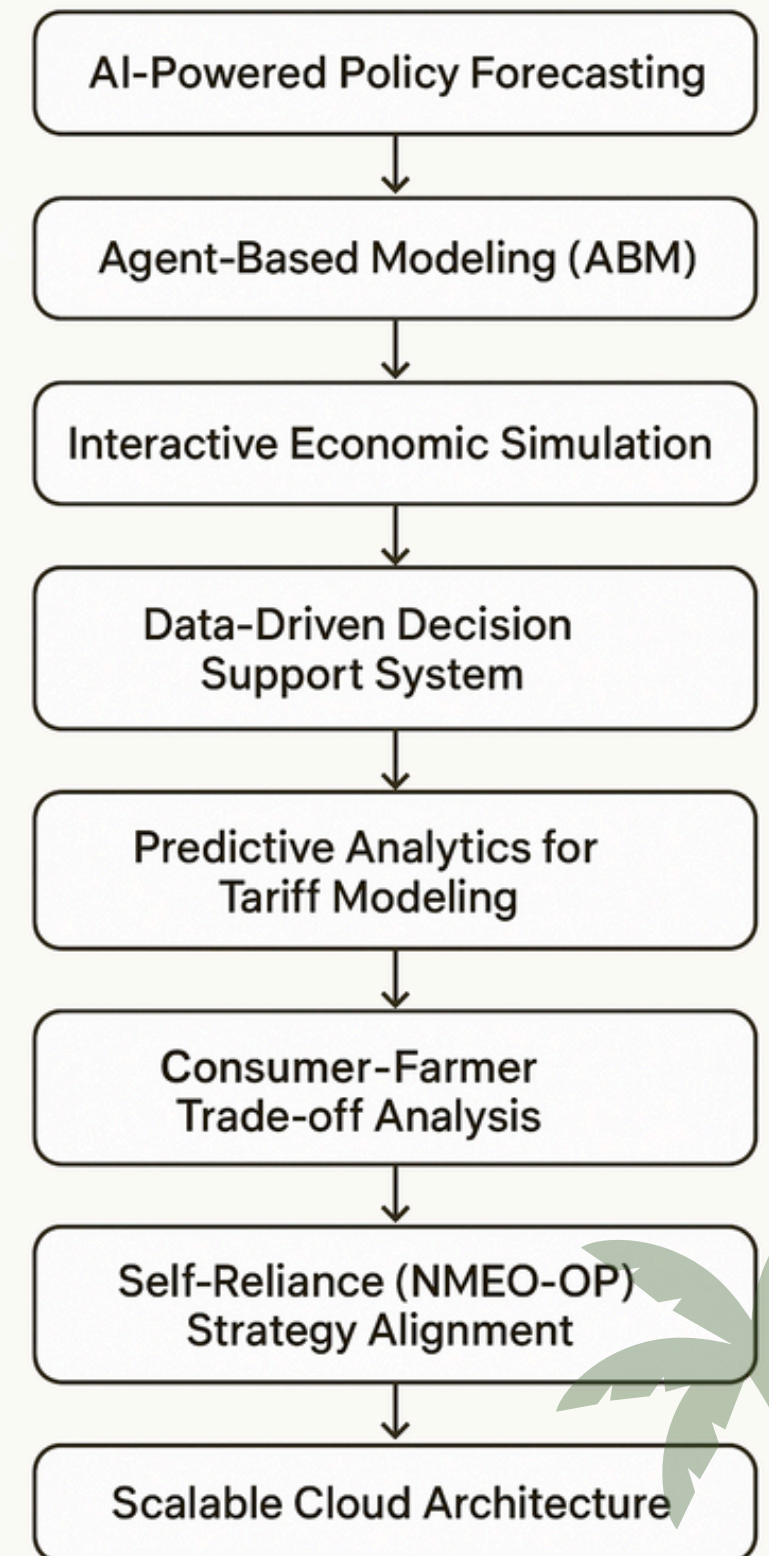
Large potential user base: Millions of crude palm oil **stakeholders benefited**.

Nationally Scalable: Expandable to other critical commodities like pulses, spices and cotton.

Long-term value: Potential integration with **state integration department** and FPO with Facilitates continuous engagement and benefit.

Economic Impact : Contributes to the **food security and import reduction cost**.

Effectivity: **High Forecast Accuracy** Value with **LSTM Model**

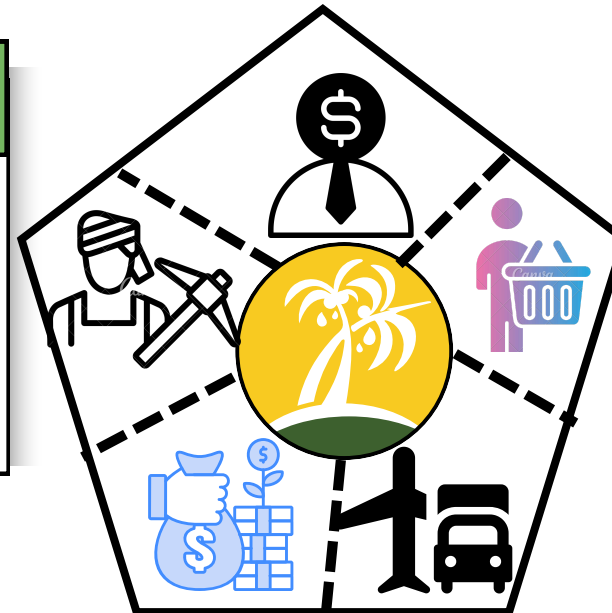


Impact on Traders

Price Transparency(Explainable AI(XAI))
Accurate Price Information
Upto Date Prediction

Impact on Farmers

- » Optimized Crop Planning
- » Reduced Market Manipulation
- » Land Expansion
- » Fair Trade Practices



Impact on Consumers

- » Reliable Supply of Product
- » Timely and Complete Price Relief
- » Budget Management
- » Forex Saving

Impact on Investors

Risk-Aware Investment
Data Driven Long-term price
Confidence
Align investment with Nation
Growth Goals

Impact on Importers

- » Reliable Long-term Trade Forecasts
- » Reduced Business Uncertainty, Better planning
- » Predicts Market Shocks, Optimizes Logistics

Benefits

Direct Alignment with National Goals:

- implementing the principles of **NMEO-OP**, 'Make in India,' and '**Atmanirbhar Bharat**'.

Fair Trade Practices :

- Ensures that the benefits of government tariffs reaches down to the cultivators.

Inflation Control :

- Helpful in Management of **Consumer Price Index(CPI)**, as edible oils are key component of Household Spending.

Food Security:

- Prediction and **control of National Food Security**.

Strengthened Digital India Initiative :

- Connecting for Trade Nationally.

Technical References & Research:

-  **Forecasting at Scale:** The foundational paper describing the Prophet time-series model used for price and volume forecasting.
-  **Prophet Documentation, Meta Open Source, 2023:** Official documentation for the Prophet Python/R library, detailing its robust features for handling seasonality and trend shifts.
-  **LSTM Models for Time Series Forecasting** The Long Short-Term Memory (LSTM) model is explicitly used in your AI Forecasting layer for high-accuracy predictions (MAPE $\approx 3.4\%$). (No single link is provided in the original text, but this search query will link to relevant research.

Data Sources & Official Documents

-  **NMEO-OP Mission Guidelines:** Provides the policy mandate and objectives our simulator is designed to support
-  **India Import-Export Database:** Provides the domestic trade statistics and real-time import data used by the system.
-  **Global Oil Crop Market Data:** Provides international price trends and production data necessary for global market forecasting.

[Our Prototype Explanation](#)

[Our Research Paper :](#)